RS Procedure 2016_037

Dec 14th

2016

This procedure describes about the how to replace the "Wear Strip" plates in RS. This is applicable to S-2000i250B, S-2000i300B, but you can use similar procedure for other model also"

[**Wear Strip replacement in Roboshot machines**]

Procedure for Replacement of the Scratched Wear Strip in ROBOSHOT S-2000*i*250B, S-2000*i*300B

General Cautions

a. Safety first.

b. Special care will be needed for handling any parts.

c. Mark on the any parts for assembling in the original position and direction when disassembling.

Required tools

- General working tools
- Four slings (same length)

  The slings with no damages are needed. (The weight of the rear platen is about 1.5 ton.)
- Four eye bolts M20

  Confirm no cracks or no bending on the male threads. (New eye bolts are recommended.)
- Oil stone

  Prepare the fine-grit oil stone.
- Penetrant CRC 5-56 or the like

  The low viscosity spray type of penetrant will be easy to use.
- Inside micrometer (for measurement of platen parallelism)
- Mold

  The mold used on the machine as frequently as possible will be needed.
- Oil-based Marker pen
- Grease gun (adopting the cartridge grease)
- Grease FS-2 (400cc cartridge for grease gun)
- LOC-TITE 242

  Medium adhesive for threads.

  Apply it to all the screw threads except the cover screws unless otherwise specified.

- The tightening torque of cap screws

  Tighten the screw in the following torque, unless otherwise specified.

Tightening torque for Cap Screw

Screw size	M6	M8	M10	M12	M14	M16
Torque (Nm)	12.3	28.9	58.8	101	162	245

Working Procedure

The temporary treatment is partially different from the permanent countermeasure.

Temporary treatment (Repair): The following cases can be practiced.

1) In the case of the light scratches or a small number of scratches on the wear strip found.
2) In the case of some emergency treatment required.

For both of cases above, the temporary treatment (recycling by grinding the damaged wear strip) will be adopted, and the permanent countermeasure will be needed again. (By consultations with the customer, the case of no permanent countermeasure needed will be excluded from the countermeasure.)

Permanent countermeasure (Replacement of new parts) The procedure of replacing a new parts will be needed additionally.

Temporary treatment (Repair activity)

(1) Mount the mold for the actual moldings, and execute the automatic die height adjustment for the rated tonnage setting.

- The mold used on the machine as frequently as possible will be needed. (Metering : OFF, Mold temperature control : OFF)
- In the case of mold temperature ON just before working, wait for the temperature going down to room temperature to start working.

(2) Measure the platen parallelism for 30 mm-opening of the mold.

- Open the mold for 30 mm in the direction of slow closing from the state of mold opening much more than 30 mm. If the mold opening gets less than 30 mm, open the mold again for much more than 30 mm, and move in the direction of closing the mold to set the state of 30 mm opening.
- Measure the platen parallelism for 30 mm-opening of the mold referring to Fig.1, and record the parallelism data. In this step, measurement and recording are required. No adjustment is needed here.

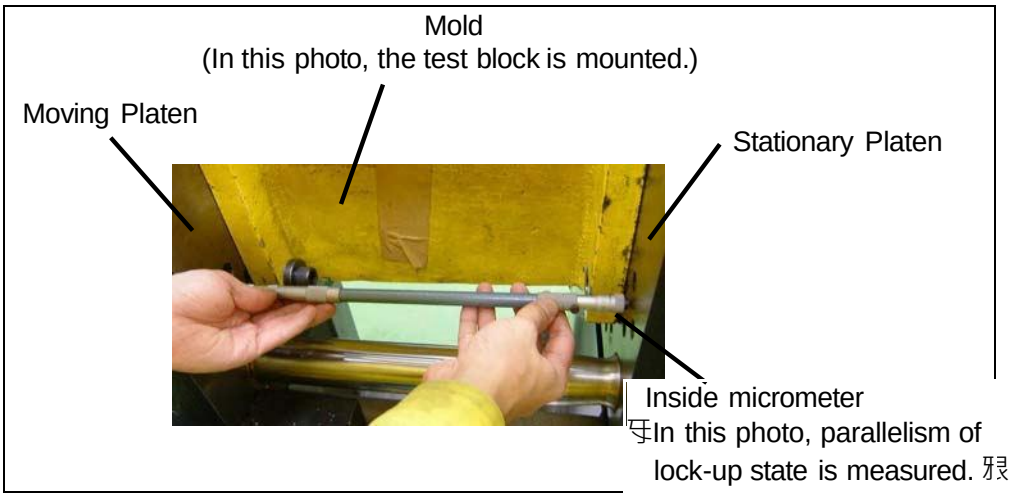

Fig.1 Parallelism measurement by Inside micrometer

(3) After dismounting the mold, turn off the main breaker.

(4) Remove the covers of the clamping unit.

- Removing the cover of stationary platen is not needed.
- Remove the covers such as the operation side gate, the non-operation side gate and the side covers preventing from attaching/detaching the wedges of MV Platen supports. And grind the wear strips by the use of oil stone.

On the step (5) to step (8) mentioned below, start working for the operation side at first. After completing the work on the operation side, work on the non-operation side.

(5) Mark on the wedge parts (on both the Stationary platen side and the Rear platen side) by the oil-based marker pen. After marking on, remove the wedges. (ref. to Fia.2(a).)

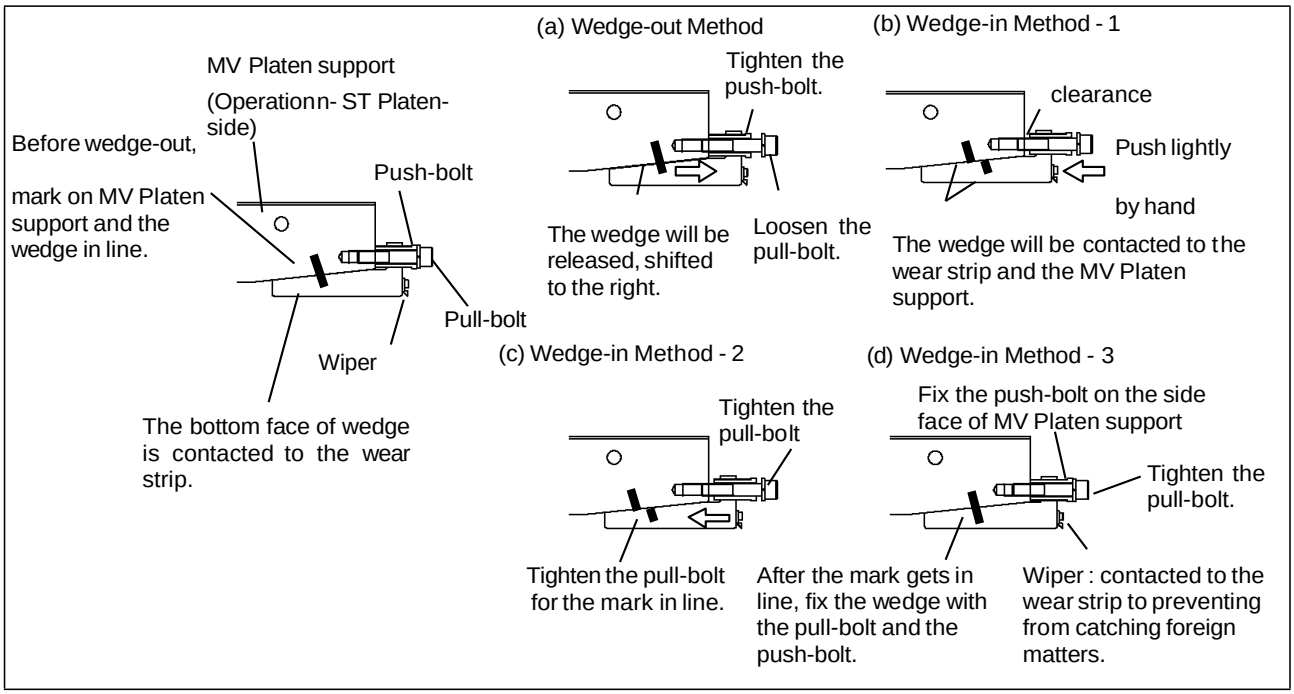

Fig.2 Wedge-in and wedge-out methods for MV Platen support

(6) Grind out the burr of scratching the bottom faces of wedge (contacting the wear strip) by the use of the oil stone.

- Penetrate a lubricant like CRC 5-56 on the oil stone in advance.

(7) Grind out the burr of scratching the wear strip by the use of the oil stone as well as (6).

- In the case of the screws for mounting the wear strip on the base preventing from grinding the wear strip, unscrew them as needed. And screw them again after completion of grinding.

(8) After completion of (6), apply the grease FS-2 on the bottom face of wedges. And mount the wedges on the MV Platen support, referring to Fig.2(b),(c),(d).

- The wiper fixed on the outside of each wedge is important to preventing from catching the foreign objects. Fit the wipers on the wear strip.

After completion of step (5) to step (8) for the operation side mentioned above, work on the non-operation side as well.

(9) After completion of step (5) to step (8) for four wedges, turn on the main breaker, and mount the same mold as step (1) for measuring the platen parallelism. And then, execute the automatic die height adjustment for the rated tonnage setting.

(10) Measure the platen parallelism on the 30 mm opening state as well as step (2).

(11) If the parallelism in step (10) is very different from the parallelism in step (2), the fine-adjustment for four wedges will be needed to get closed to the parallelism in step (2).

(12) After (11) completed, dismount the mold and put the covers on the machine.

Completion of temporary treatment

Permanent countermeasure

(1) Mount the mold for the actual moldings, and execute the automatic die height adjustment for the rated tonnage setting.

- The mold used on the machine as frequently as possible will be needed Metering: OFF, Mold temperature control: OFF
- In the case of mold temperature ON just before working, wait for the temperature going down to room temperature to start working.

(2) Measure the platen parallelism for 30 mm-opening of the mold.

- Open the mold for 30 mm in the direction of slow closing from the state of mold opening much more than 30 mm. If the mold opening gets less than 30 mm, open the mold again for much more than 30 mm, and move in the direction of closing the mold to set the state of 30 mm opening.
- Measure the platen parallelism for 30 mm-opening of the mold referring to Fig.3, and record the parallelism data. In this step, measurement and recording are required. No adjustment is needed here.

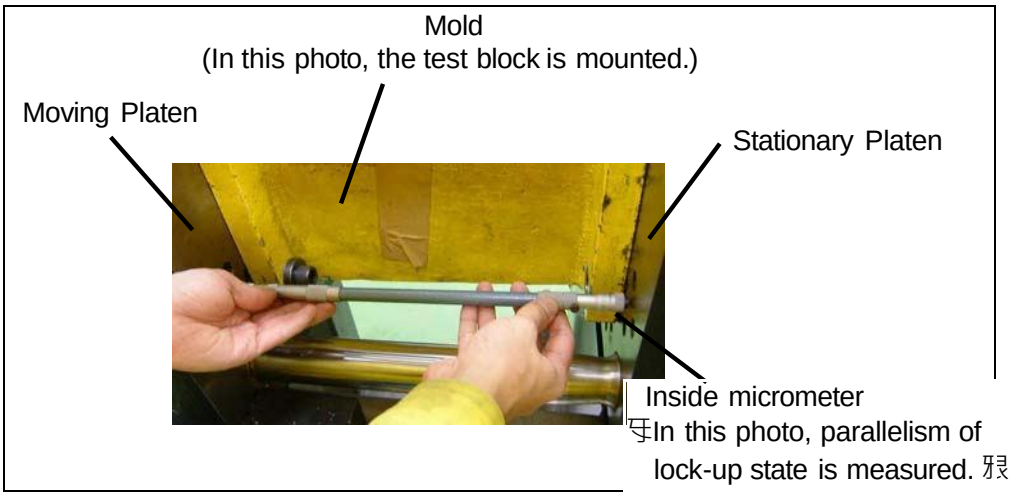

Fig.3 Parallelism measurement by Inside micrometer (same as Fig.1)

(3) Dismount the mold. And retract the rear platen closed to retract OT (between retract Over Travel point and retract Over Travel point – 5 mm) by manual.

Important caution:

Slight lifting up the rear platen with four strain-rods assembled in the rear platen will be needed in the following steps.
This step shall be done for decreasing the bending load of strain-rods and for safety.

(4) Turn off the main breaker.

(5) Remove the covers of the clamping unit.

- Removing the cover of stationary platen is not needed.
- Remove the covers such as the operation side gate, the non-operation side gate and the side covers preventing from attaching/detaching the MV Platen support units.
- Remove the top covers of the clamping unit, such as the top covers of rear platen and the toggle links. Because slight lifting up the rear platen will be needed in the following steps.

(6) Remove the MV Platen support units on the operation side and non-operation side

- Before removing the MV Platen support units, release four wedges, referring to Fig.4(a).

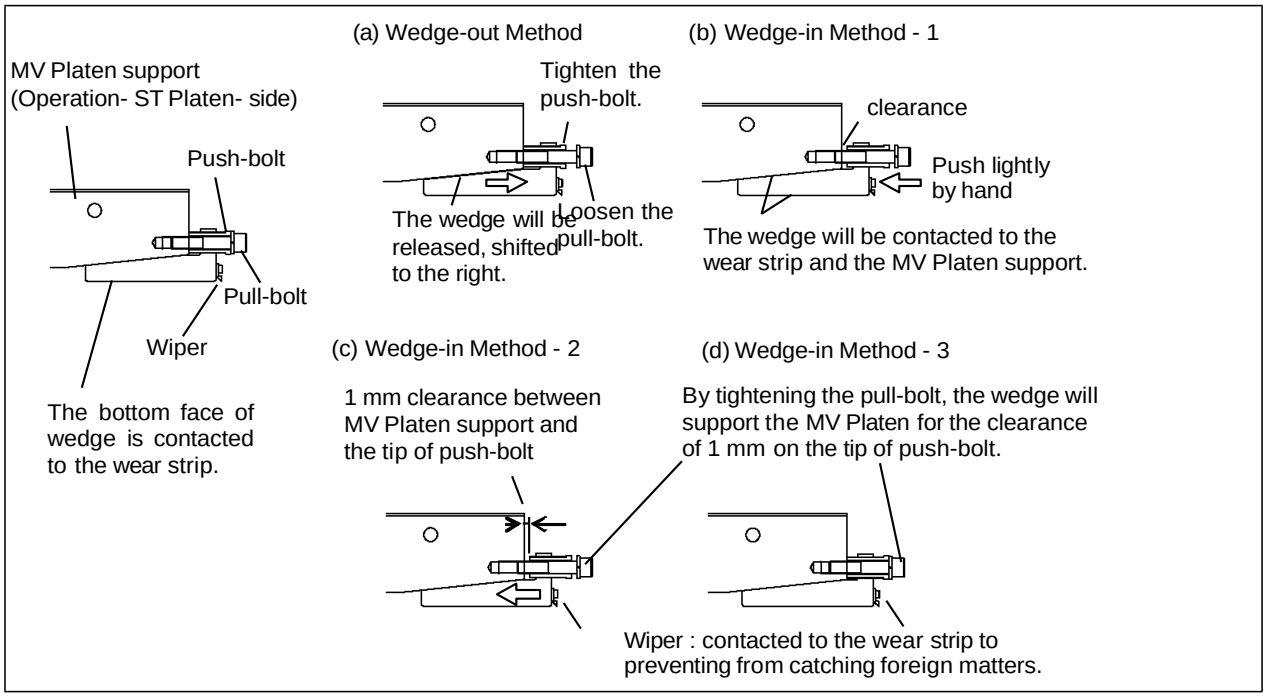

Fig.4 Wedge-in and wedge-out methods for MV Platen support

(7) Mount the eye bolts (4xM20) on the top face of rear platen and lift up the rear platen slightly (less than 3 mm) by the use of the slings and the eye bolts. After that, replace the wear strips (both on operation side and on non-operation side : 2 mm thick).

Important cautions:

- Slight lifting up the rear platen shall be needed for the reasons below.
  - Lifting up the rear platen needed only for the replacement of thin (2 mm thick) wear strips
  - The bending moment of the strain-rods as small as possible.
  - The loads acting on the crane, the slings and the eye bolts as small as possible.
- There is a danger that the slings or the eye bolts are failed by the excess lifting-up force. New slings and eye bolts or no-damage slings and eye bolts are strongly recommended.
- Do not put your fingers under the rear platen lifted up. There is a danger of serious injury.

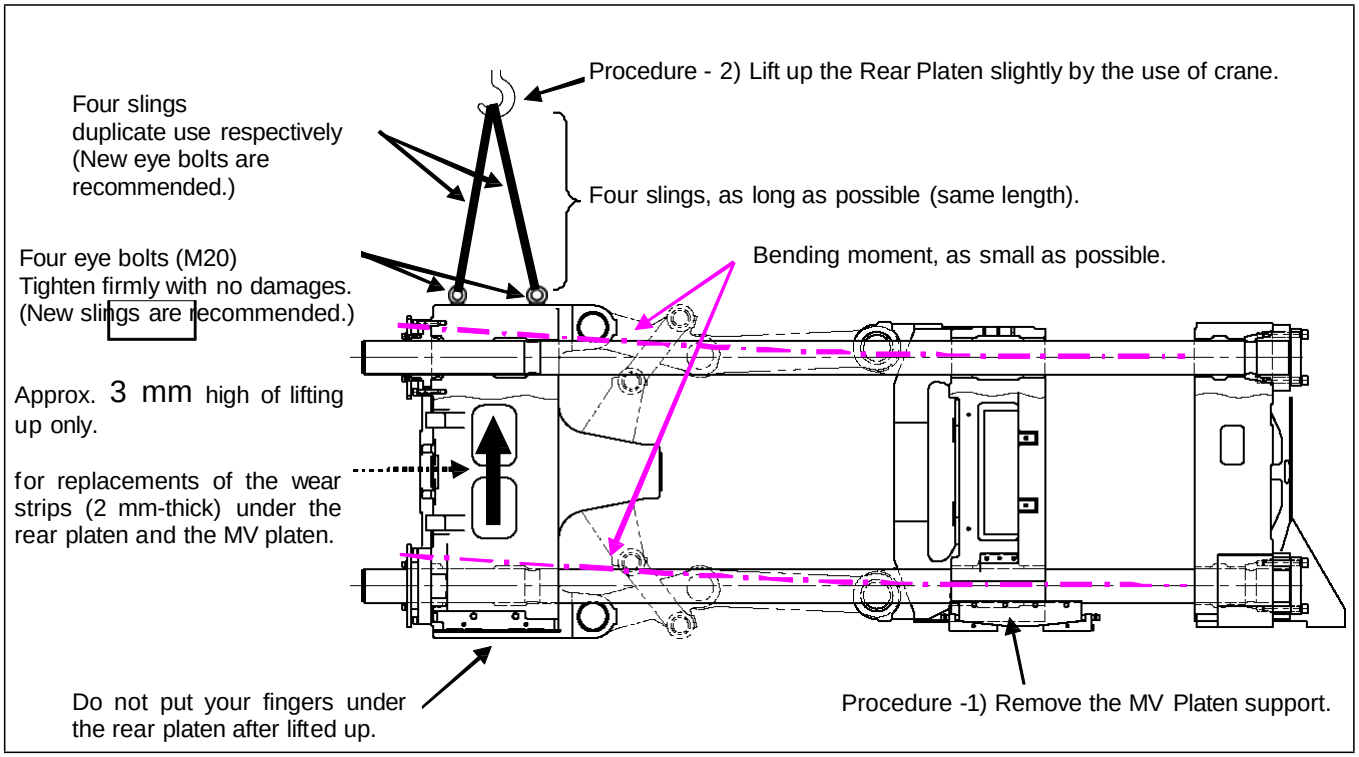

Fig.5 cautions on lifting the Rear platen

- After removing an old wear strip, grind the face of base for fixing the wear strip slightly by the use of oil stone. Penetrate a lubricant like CRC 5-56 on the oil stone in advance.
- Apply the grease FS-2 on the rear platen stroke on the new wear strips in advance.
- Just after replacement of the wear strips completed, lift down the rear platen and remove the slings and the eye bolts.

(8) Assemble new MV Platen supports (for operation side and for non-operation side) to the MV platen.

- Before assembling, confirm that no burr or no foreign objects stay on the bottom faces of four wedges (contacting the wear strips). And apply the grease FS-2 on the bottom faces.
- Before assembling, release four wedges referring to Fig.4(a).
- The MV Platen support units shall be fitted firmly on the assembling faces of MV platen.

(9) Assemble the wedges on the new MV Platen supports referring to Fig.4(b),(c),(d).

- On the every step for Fig.4(b),(c) and (c), assemble in the following order
  - ȣ Operation- and ST platen- side
  - ŋ Non operation- and ST platen- side
  - ȣ́ Non operation- and Rear platen- side
  - ʊ́ Operation- and Rear platen- side
- The wiper fixed on the outside of every wedge is important to prevent from catching foreign objects. Fit the wipers on the wear strip.

(10) Turn on the power.

(11) Mount the mold for the actual moldings, and execute the automatic die height adjustment for the rated tonnage setting.

- Before execution of the automatic die height adjustment, forward the rear platen by manual, as needed.
- mount the same mold as step (1)
- Metering: OFF, Mold temperature control: OFF

(12) Measure the platen parallelism for 30 mm-opening of the mold.

- Open the mold for 30 mm in the direction of slow closing from the state of mold opening much more than 30 mm. If the mold opening gets less than 30 mm, open the mold again for much more than 30 mm, and move in the direction of closing the mold to set the state of 30 mm opening.
- Measure the platen parallelism for 30 mm-opening of the mold referring to Fig.3.

(13) If the parallelism in step (12) is very different from the parallelism in step (2), the fine-adjustment for four wedges will be needed to get closed to the parallelism in step (2).

(14) After (13) completed, dismount the mold and put the covers on the machine.

Completion of permanent countermeasure

Common to Temporary treatment and Permanent countermeasure

After completion of working for the wear strip, modify the parameters defining the greasing interval of the central lubrication system and the greasing amount following the other instruction, as needed.

"The tie bar balance doesn't change greatly before and after replacing. However, we recommend the parallelism and the tie bar balance to be confirmed"